

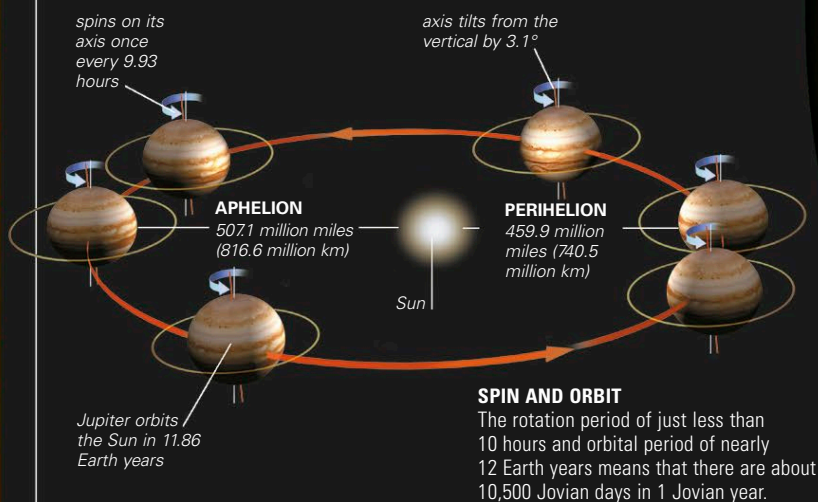
JUPITER

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JUPITER IS THE LARGEST AND MOST MASSIVE of all the planets. It has almost 2.5 times the mass of the other seven planets combined, and over 1,300 Earths could fit inside it. Jupiter bears the name of the most important of all the Roman gods (known as Zeus in Greek mythology). The planet has the largest family of moons in the solar system, its members named after Jupiter's lovers, descendants, and attendants.

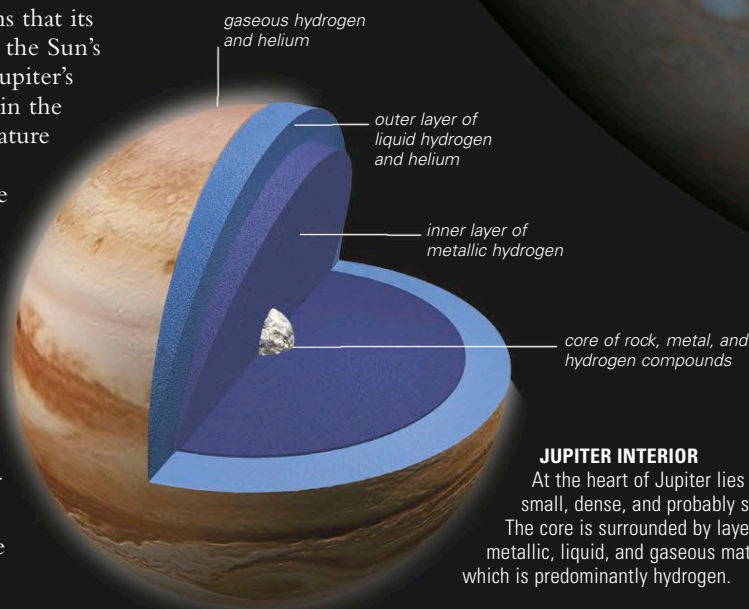
ORBIT

Jupiter is the fifth planet from the Sun. It lies approximately five times as far away as Earth, but its distance from the Sun is not constant. Its orbit is elliptical and there is a difference of 47.3 million miles (76.1 million km) between its aphelion and perihelion distances. Jupiter's spin axis tilts by 3.1° , and this means that neither of the planet's hemispheres point markedly toward or away from the Sun as it moves around its orbit. Consequently, Jupiter does not have obvious seasons. The planet spins quickly around its axis more quickly than any other planet. Its rapid spin throws material in its equatorial region outward. The result is a bulging equator and a slightly squashed appearance.



STRUCTURE

Although it is the most massive planet (318 times the mass of the Earth), Jupiter's great size means that its density is low. Its composition is more like the Sun's than any other planet in the solar system. Jupiter's hydrogen and helium is in a gaseous form in the outer part of the planet, where the temperature is about -166°F (-110°C). Closer to the center, the pressure, density, and temperature increase. The state of the hydrogen and helium changes accordingly. By about 4,350 miles (7,000 km) deep, at about $3,600^\circ\text{F}$ ($2,000^\circ\text{C}$), hydrogen acts more like a liquid than a gas. By 8,700 miles (14,000 km), at about $9,000^\circ\text{F}$ ($5,000^\circ\text{C}$), hydrogen has compacted to metallic hydrogen and acts like a molten metal. Deep inside, at a depth of about 37,260 miles (60,000 km), is a solid core of rock, metal, and hydrogen compounds. The core is small compared to Jupiter's great size but is about 10 times the mass of Earth.



JUPITER INTERIOR

At the heart of Jupiter lies a relatively small, dense, and probably solid core. The core is surrounded by layers of metallic, liquid, and gaseous material, which is predominantly hydrogen.